

10.4 Determining Appropriate Measures of Center and Variation

Lesson Introduction

 Benchmark	MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, while taking into consideration the context and any outliers.		 Learning Target	I can determine an appropriate measure of center and variation to summarize numerical data.
 Benchmark Coherence	Building On MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range. MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.	
 Essential Questions	- How do you determine which measures of center best describe a data set? - How do you determine which measures of variation best describe a data set?			
 Lesson Narrative	In this lesson, students continue to build their understanding of how to determine which measures of center and variation best describe sets of data based on real-world contexts. Emphasis should be placed on the context, not only the visual representation of the data. In the Guided Practice, students work through scaffolded prompts to determine the measures of center and variation that most accurately describe the given data sets. As the lesson progresses, students continue to interpret data sets and describe them using appropriate measures of center and variation through less scaffolded means, culminating in a Wrap-Up requiring students to interpret and describe data presented in a line plot.			
 Component Summary	10.4.1 Warm-Up (~5 min.)	Create a ratio to represent a real-world context (<i>SE p. 123</i>)	10.4.3 Try It (10-15 min.)	Explain which measures of center and variation most accurately synthesize a data set (<i>SE p. 125</i>)
	10.4.2 Guided Practice (10-15 min.)	Describe measures of center and variation to best summarize a data set (<i>SE p. 123</i>)	10.4.4 Your Turn! (10-15 min.)	Interpret a histogram to determine the appropriate measures of center and variation to describe the data (<i>SE p. 126</i>)
	10.4.5 Wrap-Up (~5 min.)	Examine the data in a line plot to choose the best measures of center and variation and explain why these measures most accurately describe the data (<i>SE p. 128</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Mean - Median - IQR - Range		N/A	
				N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may forget which measures of center and variability are best for data sets with outliers or skews.	Encourage students to refer back to the summary they previously completed when considering which measures of center and variation to use to summarize data sets.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder For the 10.4.3 Try It, consider implementing a Notice and Wonder routine before students answer question 1. As students discuss what they notice about the data, circulate to listen for students who are noticing the outliers in the data and discussing what that means for the measures of center and variation. Highlight these for the class in a whole-group discussion.	MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students connect measures of center and variation to the contexts of the data sets and their purpose in the real world. Students use context to determine which measures are most appropriate to summarize and make decisions about the data set.
 Differentiate Instruction	10.4.2 Guided Practice offers opportunities to include multiple entry points for students to access the content. To provide auditory entry points, students can work in pairs or small groups and discuss each problem. Visual entry points can be accessed through diagrams representing the data. Encourage students to draw box plots, histograms, or stem-and-leaf plots to help visualize the data before answering the questions.	
	Consider adding scaffolding questions to the 10.4.4 Your Turn!, similar to what is included in the 10.4.2 Guided Practice, to help students who are struggling to determine which measures of center and variation best describe the data sets. This adjustment will provide all students with an entry point to be able to apply their learning to drive conceptual understanding.	
Lesson Notes:		

10.4.1 Warm-Up: Between Two Numbers

Component Overview: Students work independently to create a ratio that represents the ratio of the sun's core temperature to the Earth's core temperature.



10.4.1 Warm-Up: Between Two Numbers

The temperature of the sun's core is 15 million degrees Celsius.

The temperature of Earth's core is 6000 degrees Celsius.

1. What is the ratio of core temperatures of the sun to Earth?

$$\frac{15,000,000}{6,000} = \frac{15,000}{6} = \frac{2,500}{1}$$



This Warm-Up reminds students of one of the first methods they used to compare just two numbers: ratios. Throughout this unit students will be asked to make comparisons within sets of data.

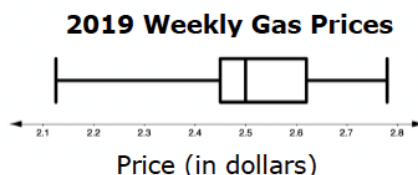
10.4.2 Guided Practice: Measures of Center and Variation

Component Overview: Students determine and explain which measures of center and variation best describe data displayed in a box plot and stem-and-leaf plot based on the context and purpose.



10.4.2 Guided Practice: Measures of Center and Variation

- The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.



- Explain which measure of center, the mean or the median, best describes the center of the data.
The median best describes the center of data. The first quartile is much larger than the others, suggesting that the data is skewed left.

- If possible, find the value of the measure you chose in Part A.
The median is 2.5 which is \$2.50 per gallon.

- Describe the variation of the weekly average price of gasoline.
Answers will vary. The middle 50% of data values, the IQR, are relatively close together. In the first quartile of data, values are more spread out. Variation is described in terms of range and interquartile range (IQR). Because the box plot does not give exact numbers for the lower and upper extreme or the quartiles, students may approximate the range to be around \$0.70 and the IQR to approximately \$0.20.

- Given your description, does the interquartile range or range better represent the variation in price?
Answers will vary. Because the box plot is skewed left, the IQR better represents the variation in price.



Students may still struggle to conclude which measures of center/variation best describes the data. To scaffold, consider asking questions like:

- Does this data set look like it is skewed or could have outliers? Why?
- When outliers are present or the distribution is skewed, which measure of center/variation usually best describes the data? Why?

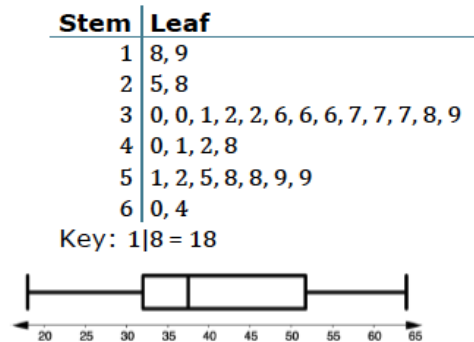


Consider providing sentence starters to provide additional support for students who may struggle to explain their thinking in writing.

- "Variation in a data set is described using _____ and _____. The _____ of the data is around _____. The _____ is approximately _____."
- "The _____ would better represent the variation in price."

10.4.2 Guided Practice: Measures of Center and Variation (continued)

2. Robin, a small business owner, is trying to determine the cost of providing a small group health care plan to her 30 employees. Her insurance agent asks her for a summary of the employees' ages, since age is a factor that affects the premium cost of insurance. The employees' ages are displayed in the stem-and-leaf plot and box plot.



Employee Ages (in years)

Complete the statements to explain which measure of center and variation Robin should provide to her insurance agent to accurately summarize the ages of her employees.

- a. The ☐ mean
☐ median best summarizes the center of the data for the agent because ...
Answers will vary. The data is approximately symmetric, so the mean and median values are close. Generally, in this case, the agent would use the mean.
- b. The ☐ IQR
☐ range best summarizes the variation of the data for the agent because ...
Answers will vary. The range should be provided since the cost of insurance depends on the ages. If only the IQR is provided, the cost may be underestimated.



Students may immediately choose IQR as the best measure of variation since they are shown a box plot. It is important to emphasize how the context of the data should inform which measure to use. In this case, the cost of the insurance premium is affected by age, so the high and low values are important to consider; thus, the range is a more helpful measure of variation to accurately estimate the insurance premium.

10.4.3 Try It: Measures of Center and Variation

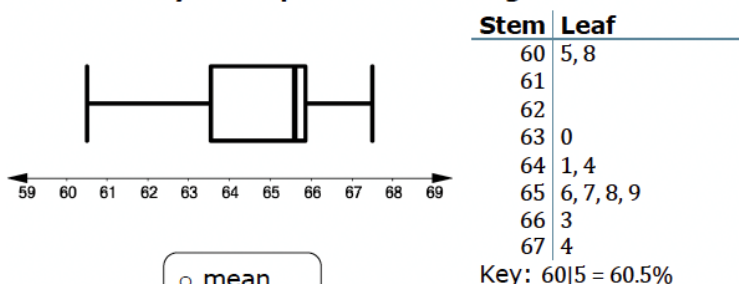
Component Overview: Students work independently or with a partner to analyze data displayed in a box plot and stem-and-leaf plot, completing statements to describe the data. Then students evaluate the data of home prices to determine which measures of center and variation best summarize the average home price in a neighborhood.



10.4.3 Try It: Measures of Center and Variation

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. His pass-completion percentages from 2009 to 2019 are shown in the box plot and stem-and-leaf plot.

Tom Brady's Completion Percentages 2009-2019



- a. The ☐ mean ☒ median best summarizes the center of the data. The ☒ mean ☐ median is not the best measure of center because it is influenced by the values ☒ 60.5% and 60.8% ☐ 66.3% and 67.4% that are far from the rest of the data values.

- b. The ☒ IQR ☐ range best summarizes the variation of the data. The ☐ IQR ☒ range is not the best measure of variation because it is affected by the value ☒ 60.5% ☐ 67.4% that is far from the rest of the data.



Notice and Wonder

It may be helpful to implement a Notice and Wonder routine before students complete question 1. This will provide multiple entry points through low stakes questioning, ensuring all students can access the material in some way. As students are discussing what they notice about the data, circulate to listen for students who are noticing the outliers in the data and discussing what that means for the measures of center and variation used to describe the distribution. Highlight these for the class in a whole-group discussion.

10.4.3 Try It: Measures of Center and Variation (continued)

2. The Rachards family is moving to Gainesville, FL and want to purchase a home in a specific area. They have a budget of \$160,000. The prices of homes in the area are shown below, in thousands of dollars.

150, 302, 201, 212, 150, 160, 140, 145

The Rachards are concerned that the homes may be out of their budget, so they ask the realtor to summarize the home prices.

- a. Which measure of center should the realtor report to the Rachards, the mean or the median? Explain your reasoning.

Answers will vary. The realtor should report the median to the Rachards because it is not affected by the extreme value of 302 thousand dollars. If the realtor gives the mean, the Rachards may think they can't afford a home in the area.

- b. Which measure of variation should the realtor report to the Rachards, the interquartile range or the range? Explain your reasoning.

Answers will vary. The realtor should report the interquartile range to the Rachards because it provides the variation of the center of the data.



Consider ways to provide multiple entry points for students to access the content in this component. To deliver auditory entry points, students can work in pairs or small groups and discuss each problem. Visual entry points can be accessed through diagrams representing the data. Encourage students to draw box plots, histograms, or stem-and-leaf plots to help visualize the data before answering the questions.

10.4.4 Your Turn!

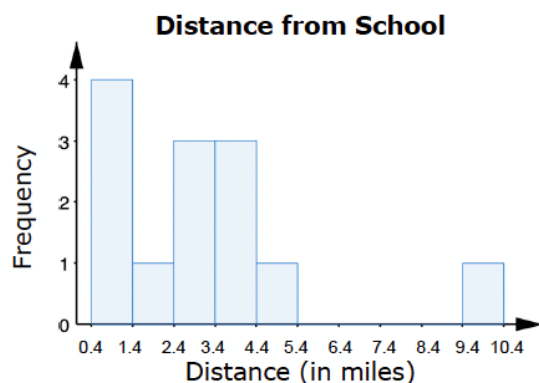
Component Overview: Students work independently to analyze data displayed in a histogram, determining which measures of center and variation best describe the data and why. Students then consider whether the measures of center and variation need to change once the outlier of the data set is removed.



10.4.4 Your Turn!

Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 10.3



1. Zane concluded that the best description of the typical distance that a student in his class lives from school is 3 miles, the value of the median.

Do you agree with Zane? Explain your reasoning.

Answers will vary. Zane's decision to use the median to describe the typical distance that a student in his class lives from school is the best because it is not affected by the outlier of 10.3.

2. Zane explained to his classmate, Shelby, that the students lived anywhere from 0.4 to 10.3 miles from school, so the range of 9.9 miles was the best summary for the spread of the data.

Shelby argued, "That's misleading, only one person lived more than 5.4 miles from school, so the interquartile range is a more accurate depiction of the spread of the distances."



Students may have trouble explaining their reasoning. If this is the case, you can scaffold the entry point by reminding them to consider the outlier(s). You may, also, encourage them to utilize the units on the horizontal axis as a means to the real-world context of the question.

10.4.4 Your Turn! (continued)

- a. Do you agree with Zane or Shelby? Explain your reasoning.

Answers will vary. Shelby's use of the interquartile range is a more accurate depiction of the spread of the distances because it is not affected by the outlier of 10.3.

- b. Under what circumstances would you agree with the other person's chosen measure of variation?

Answers will vary. If the intent was to estimate how long it would take to visit each person's house, the range would be useful since the high value would impact the time required.

3. Noelle is Zane's friend who used to live 10.3 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school.

The data set with Noelle's new distance from school, in green, is shown.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 5.5

- a. What measure of center best describes the typical distance Zane's friends live from school? Explain your reasoning.

Answers will vary. Because the data are more symmetrical, Zane should use the mean to describe the measure of center.

- b. What measure of variation best describes the spread of the data distribution? Explain your reasoning.

Answers will vary. Since the distribution is more symmetrical, the range is the best measure of variation to describe the spread of the data.



Students may struggle to explain their reasoning in question 2b. If you notice students struggling, ask questions to further scaffold like:

- When does the range/IQR best summarize the data? What does the shape look like? What would the data look like on a box plot or line plot?
- Does the range/IQR best summarize a data set with outlier? Why or why not? What purpose for analyzing the data might change this selection in this context?



Clarifications for this benchmark include recognizing whether a measure of center or measure of variation is appropriate and can be justified based on the given context or the statistical purpose. Throughout this component, students are required to explain and justify their reasoning for their responses.



Challenge students who finish early to represent the data presented using a histogram, line plot, or box plot. Then, have them explain how their representation supports their responses.

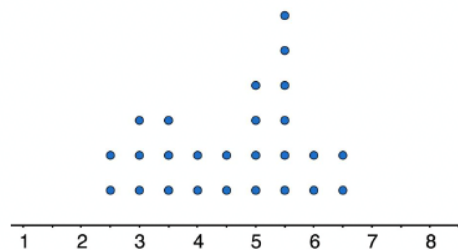
10.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students examine data in a line plot to determine measures of center and variation and explain which of these measures best describes the data.



10.4.5 Wrap-Up: Ticket Out the Door

The dot plot summarizes the hours of battery life for students' cell phones when constantly streaming videos.



Battery Life (in hours)

Mr. Funk's world history class has an upcoming field trip to Tallahassee, FL that involves a five-hour drive. The students want to be able to watch videos on their phones during the drive without charging their cell phones.

1. Determine the appropriate measure of center and variation for the battery life of the students' cell phones when streaming videos.

Answers will vary. The time the phones will be able to stream ranges from 2.5 hours to 6.5 hours. The mean is approximately 4.6, the median is 5, the mode is 5.5, the range is 4, and the IQR is 2. Because there is no extreme low or high, the mean most accurately describes the center of the data. The interquartile range best describes the spread of the data. The range is important to consider, however, since some students' phones will not last for the entire five-hour drive while streaming.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.